

Week 4 · Day 3

Feature Engineering, Cross-Validation & Model Comparison

Project Overview

The objective of this phase was to improve the predictive performance of the Adult Census Income classification model by introducing engineered features, comparing multiple machine learning algorithms through cross-validation, and identifying the most promising model and feature set for future hyperparameter tuning.

The work builds directly on the Day 1 dataset split, where the dataset was divided into:

- Training Set: 32,561 records
- Test Set: 16,281 records (kept completely sealed)

All feature engineering, preprocessing, model comparison, statistical testing, and feature selection experiments were performed exclusively on the training data to prevent information leakage. The held-out test set remains untouched and will only be used after model tuning is complete.

1. Engineered Features

Feature engineering was performed to create additional variables that capture relationships not explicitly represented in the original dataset. All engineered features were generated using only information available within each individual row, ensuring that no target leakage or cross-row information was introduced.

To evaluate the usefulness of each engineered feature before model training, a univariate predictive analysis was conducted using Mutual Information (MI). Higher MI values indicate a stronger relationship between the feature and the target variable.

Feature Dictionary

Feature	Type	Creation Rule	Justification	Mutual Information
net_capital	Continuous	capital_gain capital_loss	– Combines gains and losses into a single wealth-related measure.	0.123
is_married	Binary	marital_status begins with "Married"	Married individuals often have higher household earnings	0.102

			and financial stability.	
edu_x_hours	Continuous	education_num × hours_per_week	Captures the interaction between educational attainment and work effort.	0.090
log_capital_gain	Continuous	log1p(capital_gain)	Reduces skewness in capital gains and improves model learning.	0.083
age_bucket	Ordinal	Age grouped into six life stages	Captures non-linear income patterns across age groups.	0.062
higher_education	Binary	education_num ≥ 13	Represents attainment of Bachelor's degree or higher.	0.049
hours_bin	Ordinal	Part-time, Full-time, Overtime	Represents work-intensity categories.	0.038
has_capital_gain	Binary	capital_gain > 0	Indicates the presence of investment income.	0.030

Key Findings

The strongest individual signals came from features related to capital income and education-work interactions. In particular, net_capital, is_married, and edu_x_hours achieved the highest mutual information scores.

Although has_capital_gain produced the lowest mutual information score, it generated a large difference in target rates between groups, making it useful for tree-based models that can exploit rare but highly informative splits.

Overall, the engineered features successfully introduced additional predictive information beyond the original dataset variables.

2. Leakage-Free Preprocessing Pipeline

To ensure reproducibility and eliminate data leakage, all feature engineering and preprocessing steps were embedded inside a Scikit-Learn pipeline.

The pipeline consisted of the following stages:

Feature Generation

A `FunctionTransformer` was used to generate all engineered features during training. Because the transformation occurs inside the pipeline, features are recalculated independently within each cross-validation fold.

Numerical Features

Numerical variables were processed using:

- Median Imputation
- Standard Scaling

Median imputation handles missing values while reducing sensitivity to outliers. Standard scaling ensures numerical features operate on comparable scales for algorithms such as Logistic Regression.

Categorical Features

Categorical variables were processed using:

- Most Frequent Value Imputation
- One-Hot Encoding

One-hot encoding expanded the categorical variables into approximately 102 model-ready features.

Leakage Prevention

All preprocessing parameters, including imputation statistics and scaling values, were learned only from the training portion of each fold. During cross-validation, the validation fold remained completely unseen until evaluation.

This design guarantees that performance estimates accurately reflect real-world generalization.

3. Cross-Validated Model Comparison

Three machine learning algorithms were evaluated using Stratified 5-Fold Cross-Validation:

1. Logistic Regression
2. Random Forest Classifier
3. HistGradientBoosting Classifier

Stratified folds were used to preserve the original class distribution in every fold. Since only approximately 24% of individuals earn more than \$50K annually, F1 Score was selected as the primary evaluation metric. ROC AUC and Accuracy were also reported to provide a more comprehensive assessment.

Cross-Validation Results

Model	Accuracy	F1 Score	ROC AUC	Fit Time
Logistic Regression	0.855 ± 0.003	0.665 ± 0.007	0.910 ± 0.002	3.3 s
Random Forest	0.857 ± 0.003	0.680 ± 0.007	0.907 ± 0.002	29.7 s
HistGradientBoosting	0.873 ± 0.003	0.715 ± 0.007	0.928 ± 0.001	11.6 s

Interpretation

HistGradientBoosting consistently outperformed the other models across all evaluation metrics. It achieved the highest Accuracy, F1 Score, and ROC AUC while maintaining a relatively low training time.

Random Forest provided only a modest improvement over Logistic Regression while requiring substantially more computational resources.

Logistic Regression remained competitive despite its simplicity and continues to serve as a strong baseline model.

Cross-Validation Visualization

Boxplots of fold-level scores showed that HistGradientBoosting not only achieved the highest average performance but also demonstrated the smallest variability across folds. This indicates a stable and reliable model that is less sensitive to data partitioning.

4. Statistical Comparison and Feature Importance

Statistical Significance Testing

Because HistGradientBoosting and Logistic Regression emerged as the strongest candidates, their ROC AUC scores were compared using paired statistical tests.

Results

Comparison	Mean ROC AUC Difference	Paired t-Test	Wilcoxon Test
HistGradientBoosting vs Logistic Regression	+0.0176	t = 27.59, p = 0.00001	W = 0, p = 0.0625

Interpretation

The paired t-test produced a p-value far below the conventional significance threshold of 0.05, indicating that the performance improvement of HistGradientBoosting is highly unlikely to be due to random variation.

Although the Wilcoxon test did not reach statistical significance, this result is largely attributable to the very small sample size of five folds. Importantly, every fold favored HistGradientBoosting, supporting the conclusion that the observed improvement is meaningful.

From a practical perspective, the model achieved:

- Approximately 1.8% higher ROC AUC
- Approximately 5% higher F1 Score
- Faster training than Random Forest

These gains are substantial enough to justify selecting HistGradientBoosting as the primary model for tuning.

Feature Importance Analysis

Feature importance analysis was performed using the tree-based model and coefficient analysis was conducted for Logistic Regression.

Most Influential Engineered Features

1. edu_x_hours
2. net_capital
3. log_capital_gain

4. is_married
5. higher_education

The strong performance of edu_x_hours confirms the hypothesis that education and work effort interact to influence earning potential.

Similarly, net_capital and log_capital_gain ranked highly because investment-related income is strongly associated with higher-income individuals.

Within Logistic Regression, both log_capital_gain and has_capital_gain received meaningful coefficients, indicating that both the existence of investment income and its magnitude contribute to income prediction.

Overall, the feature importance results align closely with domain knowledge and support the validity of the engineered features.

5. Feature Selection and Dimensionality Assessment

To determine whether reducing dimensionality could improve performance or efficiency, SelectKBest feature selection based on Mutual Information was evaluated.

Results

Configuration	ROC AUC	Training Time
All Features	0.9276 ± 0.0012	11.0 s
SelectKBest (k=40)	0.9251 ± 0.0019	47.3 s
SelectKBest (k=30)	0.9243 ± 0.0019	46.0 s
SelectKBest (k=20)	0.9219 ± 0.0010	46.7 s

Interpretation

Feature selection consistently reduced predictive performance while increasing computational cost. The additional Mutual Information calculations required during each cross-validation fold introduced significant overhead.

The engineered feature set is already compact, informative, and largely free of redundancy. Therefore, removing features provided no measurable benefit.

As a result, the full feature set will be retained for future experimentation.

6. Recommendations for Hyperparameter Tuning

Based on the results of feature engineering, model comparison, statistical testing, and feature selection, the following decisions were made for Day 4 hyperparameter optimization.

Decision Area	Recommendation	Rationale
Primary Model	HistGradientBoosting	Best performance across all metrics and statistically significant improvement
Baseline Model	Logistic Regression	Fast, interpretable, and competitive baseline
Feature Set	All Original + 8 Engineered Features	Feature selection reduced performance
Primary Metric	F1 Score	Most appropriate for the class imbalance present in the dataset
Secondary Metric	ROC AUC	Provides threshold-independent evaluation
Optional Feature Removal	has_capital_gain, hours_bin	Lowest overall importance if faster training becomes necessary

Conclusion

The introduction of engineered features significantly enriched the predictive information available to the models, particularly through capital-income indicators and interaction-based features. Cross-validation demonstrated that HistGradientBoosting consistently outperformed Logistic Regression and Random Forest across Accuracy, F1 Score, and ROC AUC while maintaining reasonable computational efficiency.

Statistical testing confirmed that the performance improvement was meaningful rather than the result of random variation. Feature selection experiments further showed that the current feature set is already well optimized, making dimensionality reduction unnecessary.

Based on these findings, HistGradientBoosting trained on the complete feature set will serve as the primary candidate for Day 4 hyperparameter tuning, while Logistic Regression will remain as an interpretable benchmark model.

Visualizations:



